Unit 6.01: Linear Regression & OLS

Q1. A fitted simple linear regression model is hat{y} = 7x + 18 . What is the predicted target value hat{y} when the input feature is x = 4 ?

[x] A: 46
[ ] B: 49
[ ] C: 42
[ ] D: 28

Solution: Substitute x = 4 into the regression equation: hat{y} = 74 + 18 = 28 + 18 = 46 .

Q2. In a multiple linear regression model, the Sum of Squared Errors is text{SSE} = 20 and the Total Sum of Squares is text{SST} = 100 . What is the coefficient of determination R^2 ?

[x] A: R^2 = 0.80
[ ] B: R^2 = 0.20
[ ] C: R^2 = 0.90
[ ] D: R^2 = 0.25

Solution: The coefficient of determination measures the proportion of variance explained: R^2 = 1 - frac{text{SSE}}{text{SST}} = 1 - frac{20}{100} = 1 - 0.20 = 0.80 .

Q3. In a multiple linear regression model, the Sum of Squared Errors is text{SSE} = 40 and the Total Sum of Squares is text{SST} = 200 . What is the coefficient of determination R^2 ?

[x] A: R^2 = 0.80
[ ] B: R^2 = 0.20
[ ] C: R^2 = 0.90
[ ] D: R^2 = 0.25

Solution: The coefficient of determination measures the proportion of variance explained: R^2 = 1 - frac{text{SSE}}{text{SST}} = 1 - frac{40}{200} = 1 - 0.20 = 0.80 .

Q4. In a multiple linear regression model, the Sum of Squared Errors is text{SSE} = 60 and the Total Sum of Squares is text{SST} = 300 . What is the coefficient of determination R^2 ?

[x] A: R^2 = 0.80
[ ] B: R^2 = 0.20
[ ] C: R^2 = 0.90
[ ] D: R^2 = 0.25

Solution: The coefficient of determination measures the proportion of variance explained: R^2 = 1 - frac{text{SSE}}{text{SST}} = 1 - frac{60}{300} = 1 - 0.20 = 0.80 .

Q5. In a multiple linear regression model, the Sum of Squared Errors is text{SSE} = 80 and the Total Sum of Squares is text{SST} = 400 . What is the coefficient of determination R^2 ?

[x] A: R^2 = 0.80
[ ] B: R^2 = 0.20
[ ] C: R^2 = 0.90
[ ] D: R^2 = 0.25

Solution: The coefficient of determination measures the proportion of variance explained: R^2 = 1 - frac{text{SSE}}{text{SST}} = 1 - frac{80}{400} = 1 - 0.20 = 0.80 .

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Q6. In a multiple linear regression model, the Sum of Squared Errors is $\text{SSE} = 100$ and the Total Sum of Squares is $\text{SST} = 500$. What is the coefficient of determination R^2?

[x] A: $R^2 = 0.80$
[ ] B: $R^2 = 0.20$
[ ] C: $R^2 = 0.90$
[ ] D: $R^2 = 0.25$

Solution: The coefficient of determination measures the proportion of variance explained: $R^2 = 1 - \frac{\text{SSE}}{\text{SST}} = 1 - \frac{100}{500} = 1 - 0.20 = 0.80$.

Q7. In a multiple linear regression model, the Sum of Squared Errors is $\text{SSE} = 120$ and the Total Sum of Squares is $\text{SST} = 600$. What is the coefficient of determination R^2?

[x] A: $R^2 = 0.80$
[ ] B: $R^2 = 0.20$
[ ] C: $R^2 = 0.90$
[ ] D: $R^2 = 0.25$

Solution: The coefficient of determination measures the proportion of variance explained: $R^2 = 1 - \frac{\text{SSE}}{\text{SST}} = 1 - \frac{120}{600} = 1 - 0.20 = 0.80$.

Q8. For a bivariate feature dataset, $\text{Cov}x, y = 12$, $\text{Var}x = 4$, $\bar{x} = 5$, and $\bar{y} = 25$. Compute the OLS regression slope $\hat{\beta}_1$ and intercept $\hat{\beta}_0$.

[ ] A: $\hat{\beta}_1 = 0.3333333333333333,$;
[x] B: $\hat{\beta}_1 = 3,$; $\hat{\beta}_0 = 10$
[ ] C: $\hat{\beta}_1 = 3,$; $\hat{\beta}_0 = 25$
[ ] D: $\hat{\beta}_1 = 4,$; $\hat{\beta}_0 = 5$

Solution: The OLS slope is $\hat{\beta}_1 = \frac{\text{Cov}x, y}{\text{Var}x} = \frac{12}{4} = 3$. The intercept is $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x} = 25 - 35 = 25 - 15 = 10$.